

AI Tool Usage Documentation

Tool used: Claude (Anthropic)

Purpose: Getting guidance on EDA requirements and understanding hypothesis testing for the DSA210 project phase 3 (14 april)

Summary of Interaction:

I shared my existing data quality pipeline code with Claude and asked whether my visualizations were sufficient for the EDA requirement. Claude explained that my current plots (close price time series, log return bar chart, rolling volatility) only showed how the data changes over time, but did not constitute a full EDA. Claude pointed out that a proper EDA should also include a histogram to analyze the distribution of returns, a box-plot for visual outlier detection, a scatter plot to examine relationships between variables, and descriptive statistics such as mean, median, skewness, and kurtosis printed numerically.

I then asked Claude what hypothesis testing is and how it works conceptually. Claude explained that a hypothesis test starts with a null hypothesis which always assumes no effect or no difference, and an alternative hypothesis which assumes the opposite. The p-value is then compared against a significance level α , typically 0.05, to decide whether to reject the null hypothesis. Claude also explained the test selection logic from the DSA210 flowchart: since the population variance is unknown in my case, the t-test family is appropriate; one sample compared against a constant calls for a one-sample t-test; two independent groups call for a two-sample t-test; and three or more groups call for a one-way ANOVA.

After understanding the concept, I asked Claude where and how to apply these tests in my existing code. Claude explained that since my code already computes log returns and splits the data into periods, the hypothesis tests should be placed after the data cleaning and return calculation steps. Claude specified that the one-sample t-test should be applied directly on the returns array using `scipy.stats.ttest_1samp`, the two-sample t-test should compare the first and second halves of the returns series using `scipy.stats.ttest_ind`, and the ANOVA should be applied across three equal time periods using `scipy.stats.f_oneway`. Claude also noted that a Levene test should be run before the two-sample t-test to check whether equal variance can be assumed.

During implementation, we also resolved a missing module error by installing `scipy`, and fixed a `matplotlib` rendering issue on macOS by switching to the `Agg` backend and saving plots as PNG files that open automatically after the script runs.

Prompts used:

"Does my current visualization count as EDA, or am I missing something?"

"What is hypothesis testing and how is it calculated conceptually?"

"Which hypothesis tests should I use for my stock return data and why?"

"Where exactly in my code should I add these tests and which functions should I use?"

Outputs used: EDA block (histogram, box-plot, scatter plot, descriptive statistics) and hypothesis testing block (three tests with visualizations) were added to the existing pipeline code based on Claude's guidance.